THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · EASY LEVEL CHAPTER TEST SET 1

Chapter 11: **Electricity** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Electric current is the rate of flow of 1

(a) mass **(b)** charge **(c)** energy **(d)** heat

2. The resistance of a wire depends on 1

(a) the time **(b)** the current only **(c)** the voltage only **(d)** its length

3. Two resistors of 5 ohm each in series have a total of 1

(a) 2.5 ohm **(b)** 10 ohm **(c)** 25 ohm **(d)** 5 ohm

4. Which of these is an insulator? 1

(a) Copper **(b)** Iron **(c)** Aluminium **(d)** Rubber

5. Thicker wires have 1

(a) the same resistance **(b)** infinite resistance

(c) less resistance **(d)** more resistance

6. Two resistors of 4 ohm each in parallel have a total of 1

(a) 2 ohm **(b)** 16 ohm **(c)** 4 ohm **(d)** 8 ohm

7. The instrument used to measure current is the 1

(a) thermometer **(b)** ammeter

(c) galvanometer only **(d)** voltmeter

8. A longer wire of the same material and thickness has 1

(a) the same resistance **(b)** no resistance

(c) less resistance **(d)** more resistance

9. In series, if one bulb fuses, the others 1

(a) glow brighter **(b)** glow dimmer **(c)** stop glowing **(d)** are unaffected

10. The instrument used to measure potential difference is the 1

(a) barometer **(b)** ammeter **(c)** ohmmeter only **(d)** voltmeter

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Define electric current. 2

12. Define the resistance of a conductor. 2

13. What is the formula for the equivalent resistance in series? 2

14. What is the SI unit of charge? Give its value in terms of electrons. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Find the charge flowing in 5 s through a wire carrying a current of 2 A. 3

16. List three factors on which the resistance of a wire depends. 3

17. Convert 1 kWh into joules. 3

OR

A current of 0.2 A flows through a resistor of 10 ohm. Find the potential difference. 3

18. Find the resistance of a series connection of 3 ohm, 4 ohm and 5 ohm. 3

OR

State three advantages of parallel connection in homes. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. State Ohm's law. Draw (describe) the V-I graph and explain what its slope represents. 5

OR

A bulb rated 100 W at 220 V is used for 10 hours. Find the energy consumed in kWh and the current drawn. 5

20. Explain the factors on which the resistance of a conductor depends. 5

OR

Define electric power. Derive P = V I = I^2 R = V^2 / R. 5

END OF QUESTION PAPER — Total: 40 Marks

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